GLA University Mathura,

B.Tech: II Year, 4th semester [M.E., C.S&E.C.]

Ethics & Values (AHE: 201)

Marks: 20 Time: 1.30 Hours

There are three sections in this question paper. Section A consists of very short answer type questions, while section B & C consist of descriptive answer type questions. Attempt questions from all sections.

Section A

1) Write short notes on each of the following :(Word Limit: 30 words, Marks: 1 mark each)

a) Values
b) Ethics
c) Agency
d) Rationality
e) Three conditions of Moral responsibility
f) Three Senses of Moral Responsibility
g) Applied Ethics
h) Professional Ethics

Section B

2) Answer any two of the following :(Word Limit: 150 words, Marks: 2 marks each)

a) Explain the idea of moral person as a natural kind.

b) What do you mean by the Engineering Ethics?

c) What do you mean by the moral responsibility?

d) Why do we need ethics in our life?

Section C

3) Answer any two of the following :(Word Limit: 250 words, Marks: 4 marks each)

a) Discuss the case of Willy Just (1.1). Put yourself in the position of Will Just. How would you have responded to the commission?

b) Explain the concept of Moral-Person.

c) Discuss the case of Compute game (1.2). How would you respond to the case as an engineer?

GLA University, Mathura
First Mid-Term Examination, 2011-12
Course: B.Tech. (CSE), II-Year, IV-Sem.
Database Management Systems (ITE221)

Uni. Roll No:

Time: 1.5 Hrs. **Total Marks: 20**

Notes:

1. All parts of a question (a, b, etc.) should be answered at one place.
2. Answer should be brief and to-the-point and be supplemented with neat sketches.
3. Any missing or wrong data may be assumed suitably giving proper justification.
4. Figures on the right-hand side margin indicate full marks.

Q1. Attempt all. **.5x16=8**

1. An advantage of the database management approach is
   (A) Data is dependent on programs (B) Data redundancy increases.
   (C) Data is integrated and can be accessed by multiple programs. (D) None of the above.
2. In the relational modes, cardinality is termed as:
   (A) Number of tuples. (B) Number of attributes.
   (C) Number of tables. (D) Number of constraints.
3. Relational calculus is a
   (A) Procedural language. (B) Non- Procedural language.
   (C) Data definition language. (D) High level language.
4. DML is provided for
   (A) Description of logical structure of database.
   (B) Addition of new structures in the database system.
   (C) Manipulation & processing of database.
   (D) Definition of physical structure of database system.
5. In the architecture of a database system external level is the
   (A) Physical level. (B) Logical level. Conceptual level (D) View level.
6. In a Hierarchical model records are organized as
   (A) Graph (B) List (C) Links (D) Tree
7. In case of entity integrity, the primary key should be
   (A) Not Null (B) Null (C) both Null & not Null (D) any value.
8. Conceptual design
   (A) is a documentation technique.
   (B) needs data volume and processing frequencies to determine the size of the database.
   (C) involves modelling independent of the DBMS.
   (D) is designing the relational model.
9. E-R model uses this symbol to represent weak entity set?
   (A) Dotted rectangle (B) Diamond (C) Doubly outlined rectangle (D)None of these
10. Key to represent relationship between tables is called
    (A) Primary key (B) Secondary Key (C) Foreign Key (D) None of these
11. In a relation
    (A) Ordering of rows is immaterial (B) No two rows are identical
    (C) (A) and (B) both are true (D) None of these.
12. Which of the following operations need the participating relations to be union compatible?
    (A) UNION (B) INTERSECTION (C) DIFFERENCE (D) All of the above
13. The full form of DDL is
    (A Dynamic Data Language (B) Detailed Data Language
    (C) Data Definition Language (D) Data Derivation Language

14. Which are the two ways in which entities can participate in a relationship?
    (A) Passive and active (B) Total and partial
    (B) Simple and Complex (D) All of the above
15. A set of possible data values is called
    (A) Attribute (B) Degree (C) Tuple (D) Domain
16. Consider the join of a relation R with relation S. If R has m tuples and S has n tuples, then the maximum size of join is:
    (A) mn (B) m+n (C) (m+n)/2 (D) 2(m+n)

Q2. Attempt any two. **2x2=4**

A. Discuss the three level architecture of database system.
B. What do you mean by data independence? Explain different type of data independence.
C. Explain the Tuple Relational Calculus.

Q3. Attempt any two. **4x2=8**

A. Distinguish between a file processing system and a database management system.
B. A database is to be constructed to keep track of the teams and games of a sport league. A team has a number of players, not all of whom participate in each game. It is desired to keep track of the players participating in each game of each team and the result of the game. Create an ER diagram, completely with attributes, keys and constraints, for the above description. Transform ER Diagrams into Relational Schema.
C. Consider the database given below, where primary keys are underlined. Give an expression in relation algebra to express each of the following queries:
DEPARTMENT (Deptno, Deptname, Location)
EMPLOYEE (Emp_No, Emp_Name, Designation, Date_Of_Join, Salary, Dept_No)

i. Display the employee name and their salary.
ii. List the departments that are located in Delhi.
iii. List the name of employee who is getting their salary in between 20000 and 35000.
iv. List the Departments that do not have any employees.
v. Display the employee name and their department name.
vi. Display the Employees (name) who are located at 'Agra'.
vii. Display the number of employees in each department.
viii. Insert a record in department table with following data:
Deptno: 50, Deptname: 'PRODUCTION', Location: 'Indore'

B.Tech (CSE) II Year, IV Semester
First Mid Term Examination – 2012
Introduction to Microprocessor-CSE222

Time: 1 Hr 30 Mins

MM: 20

Notes:-

1) Answer all questions, **Part A** is **compulsory**, **Any two** from Part B and **Any two** from Part C.
2) All parts of a question (a, b, etc.) should be answered at one place.
3) Answer should be brief and to-the-point and be supplemented with neat sketches.
4) Any missing or wrong data may be assumed suitably giving proper justification.
5) Figures on the right-hand side margin indicate full marks.

Part A

Q1) Attempt following questions. **[0.5x16=08]**

i) Which of the following is true with respect to EEPROM?
A) Contents can be erased byte wise only.
B) Contents of full memory can be erased together.
C) Contents can be erased using ultra violet rays
D) Contents cannot be erased

ii) Ready pin of a microprocessor is used
A) To indicate that the microprocessor is ready to receive inputs.
B) To indicate that the microprocessor is ready to receive outputs.
C) To introduce wait states.
D) To provide direct memory access

iii) Which of the following statement is true?
A) The group of machine cycle is called a state.
B) A machine cycle consists of one or more instruction cycle.
C) An instruction cycle is made up of machine cycles and a machine cycle is made up of number of states.
D) None of the above

iv) The no. of address lines required to address a memory of size 32 K is
A) 15 lines B) 16 lines
C) 18 lines D) 14 lines

v) In a DMA write operation the data is transferred
A) From I/O to memory. B) From memory to I/O.
C) From memory to memory. D) from I/O to I/O.

vi) Which type of JMP instruction assembles if the distance is 0020 H bytes?
A) near. B) far.
C) short. D) none of the above.

vii) Pseudo instructions are basically
A) false instructions.
B) instructions that are ignored by the microprocessor.
C) assembler directives.
D) instructions that are treated like comments

viii) Direction flag is used with
A) String instructions. B) Stack instructions.
C) Arithmetic instructions. D) Branch instructions

ix) These are two ways in which a microprocessor can come out of Halt state.
A) When hold line is a logical 1.
B) When interrupt occurs and the interrupt system has been enabled.
C) When both (A) and (B) are true.
D) When either (A) or (B) are true.

x) SD RAM refers to
A) Synchronous DRAM B) Static DRAM
C) Semi DRAM D) Second DRAM

xi) Itanium processor of Intel is a
A) 32 bit microprocessor. B) 64 bit microprocessor.
C) 128 bit microprocessor. D) 256 bit microprocessor.

xii) EPROM is generally erased by using
A) Ultraviolet rays B) infrared rays
C) 12 V electrical pulse D) 24 V electrical pulse

xiii) 8088 microprocessor has
A) 16 bit data bus B) 4 byte pre-fetch queue
C) 6 byte pre-fetch queue D) 16 bit address bus

xiv) Which microprocessor pins are used to request and acknowledge a DMA transfer?
A) reset and ready B) ready and wait
C) HOLD and HLDA D) None o these

xv) The first modern computer was called_____.
A) FLOW-MATIC B) UNIVAC-I
C) ENIAC D) INTEL

xvi) A Bus cycle is equal to how many clocking periods
A) Two B) Three
C) Four D) Six

Part B

Q2) Attempt any two of the following. **[2x2=04]**

a) Draw and explain the simple block diagram of microprocessor based system?
b) What are the functions of different types of buses used in microprocessor?
c) Define opcode and operand with respect to an instruction?

Part C

Q3) Attempt any two of the following. **[4x2=08]**

a) List the components of computer and explain each in brief. What is the difference between a microprocessor and microcontroller?
b) Give the format of flag register in 8085. Explain each flag with example?
c) Draw the pin diagram of 8085 and specify the function and direction of information flow of address bus, data bus and control bus?

GLA University, Mathura

I-Mid Term Examination, 2011-12

Course: - B.Tech (CS) II-Year, IV-Sem.

Subject: -Computer network (CSE-224) **Uni. Roll No:-**

Time:-90 Minutes **Total Marks:-20**

Notes:-

1) Answer all questions, Part A is **compulsory, Any Two** from Part B and **Any Two** from Part C.
2) All parts of a question (a, b, etc.) should be answered at one place.
3) Answer should be brief and to-the-point and be supplemented with neat sketches.
4) Any missing or wrong data may be assumed suitably giving proper justification.
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Part-A

I. Attempt all questions. **8**

Q1: Choose the correct answer for each of the following. **(1/2*12)**

i) Frequency of failure and network recovery time after a failure is measures of the _____ of a network.

a) performance b) reliability c) security d) Multiplexer

ii) _____is a mechanism used by hosts and gateway to send notification of datagram problem back to sender..

a) RARP b) ARP c) ICMP d) IGMP

iii) In a_____, each device has a dedicated point to point link to all others device.

a) Star Topology b) Mesh Topology c) Ring Topology d) Bus Topology

iv) The Internet model (TCP/IP model) consists of _____ layers.

a) Three b) Five c) Seven d) Six

v) On the incoming side of the internetwork, as the data packet moves from the lower to the upper layers, headers are _____.

a) added b) subtracted c) modified d) rearranged

vi) These can impair a signal on a network.

a) noise b) distortion c) attenuation d) all of them

vii) If the bandwidth of a signal is 5 kHz and the lowest frequency is 52 kHz, what is the highest frequency of the bandwidth while the signal is in transit along the media?

a) 57 kHZ b) 47 kHZ c) 62 kHZ d) 42 kHZ

viii) As frequency increases, the period _____.

a) decreases b) increases c) doubles d) frequency

ix) Instead of referring to digital signals as having a period and frequency when measuring a signal's bandwidth, one should refer to:

a) byte rate and byte interval b) bit rate and bit interval c) range and force d) all of above

x) Which multiplexing technique transmits digital signals?

a) FDM b) WDM c) TDM d) Both a and c

xi) In QAM, both phase and _____ of a carrier frequency are varied.

a) amplitude b) frequency c) bit rate d) baud rate

xii) If the baud rate is 400 for 4-PSK signals, the bit rate is _____ bps.

a) 4oo b) 800 c) 1600 d) 200

Q2: State True/False **(1/2*4)**

i. The Shannon capacity is a formula used to determine the theoretical data rate for a noiseless channel.

ii. Internet protocol (IP) is connectionless and unreliable protocol.

iii. Coaxial cable can carry signals of higher frequency ranges than twisted-pair cable can.

iv. Address resolution protocol (ARP) is use to map a physical address into logical address.

Part-B

II. Attempt any two **(2*2)**

Q1: Differentiate between the following:

i. Base band and broad band transmission

ii. Sky propogation and line-of- sight propogation

Q2: Discus the different level of addressing used in the internet employing the TCP/IP network model?

Q3: We have a channel with a 1-MHz bandwidth. The SNR for this channel is 63. What is the appropriate bit rate and signal level?

Part-C

III. Attempt any two **(4*2)**

Q1: Answer the following

a) Draw the graph of manchester and differential manchester scheme for the following data stream-

i. 11010010

ii. 01101100

b) Name the advantages of optical fiber over twisted pair and coaxial cable?

Q2: Answer the following.

a) Compare and contrast a datagram network and virtual circuit network? Explain why there may be greater delay in a datagram network than in a virtual-circuit network.

b) A path in digital circuit switched network has a data rate of 10 Mbps. The exchange of 1000 bits required for setup and tear down phases. The distance between two parties is 500 Km. what is the total delay, if 10000 bits of data are exchanged during data transfer phase?

Q3: What is OSI model? Briefly explain the function of different layers in OSI model?

GLA University, Mathura
I-Mid Term Examination, 2011-12

Course: - B.Tech. **II-Year, IV-Sem.**

Subject: - Computer Graphics (CSE 223) **Uni. Roll No:-**

Time:-90 Minutes **Total Marks:-20**

Notes:-

1) Answer all questions, Section A is **compulsory**, **Any Two** from Section B and **Any Two** from Section C.
2) All parts of a question (a, b, etc.) should be answered at one place.
3) Answer should be brief and to-the-point and be supplemented with neat sketches.
4) Any missing or wrong data may be assumed suitably giving proper justification.
5) Figures on the right-hand side margin indicate full marks.

PART-A

Objective Questions: (0.5*16 = 08)

I. Computer graphics uses_____ to make images.

A. Hardware and Software B. Hardware only C. Software Only D. None of these

II. GKS stands for:

A. Graphics Kernel Server B. Graphics Kernel Specification

C. Geographical Kernel System D. Graphics Kernel System

III. A computer with 32-bit word has:

A. 32 bytes /word B. 4 bytes/word C. 8 bytes/word D. 2 bytes/word

IV. Plasma display can be:

A. AC B. DC C. Hybrid AC/DC activated D. All of these

V. The Region of a display device into which a window a mapped is called

A. Windowing B. Frame C. Viewport D. image

VI . Liange- Barsky Line clipping algorithm is bases on :?

A. Region Code B. Out codes C.Parametric D. Digital Differential

VII. Boundary Fill algorithm is the simplest_____filling algorithm

A. Line B Volume C. Area D. All of these

VIII. In the Cohen-Sutherland line clipping algorithm, if both out codes are 0, the line lies entirely in the clip window and is thus, trivially:

A. Rejected B. Accepted C. Truncated D. Disappeared

IX . DDA Stands for?

A. Digital Differential Algorithm B. Digital Differential Adopter C. Digital Differential Analyzer D. None

X . If a system has Resolution **640 x 480**, total number of pixels on the screen will be?

A. 157200 B. 307200 C. 307012 D. 480 x640/2

Fill in the blanks Questions:

XI. Beam Penetration Method can produce maximum colors_____

XII. A Shadow Mask Method has _____ Electron guns of colors_____.

True and False Questions :

XIII. Plasma Panel is Non-Emissive type of display.

XIV. LED is Emissive type of display.

XV. In Raster scan system scan conversion is not required.

XVI. Random scan systems have frame buffer to store the picture.

PART-B

I. What is Frame Buffer? A Standard TV has **480** scan lines. If the aspect ratio is **¾**, what is the capacity of frame buffer needed if 2 bits per pixel are used? (2)

II. Write the advantages of Bresenham's line drawing algorithm. Using DDA line drawing algorithm find the successive points that will be plotted between **(5, 10) to (15, 30)** (2)

III. Distinguish between Flood Fill and Boundary Fill Algorithms. (2)

PART-C

I. Write Cohen Sutherland algorithm for Line Clipping. Use Liange Barsky line clipping algorithm to clip the line **P1(-15,-30) to P2(30,60)** against the window having diagonally opposite corners as **(0,0)** and **(15,15).** (4)

II. Write Midpoint Circle Drawing algorithm and execute the same for radius **r =10** and centered at **(0, 0).** (4)

III. Find the normalized Transformation N that maps a window whose lower left corner is at **(0,0)** and upper right corner is at **(2,4)** on to a viewport that has lower left corner at **(-1,-1)** and upper right corner **(-1/2,-1/2).** (4)